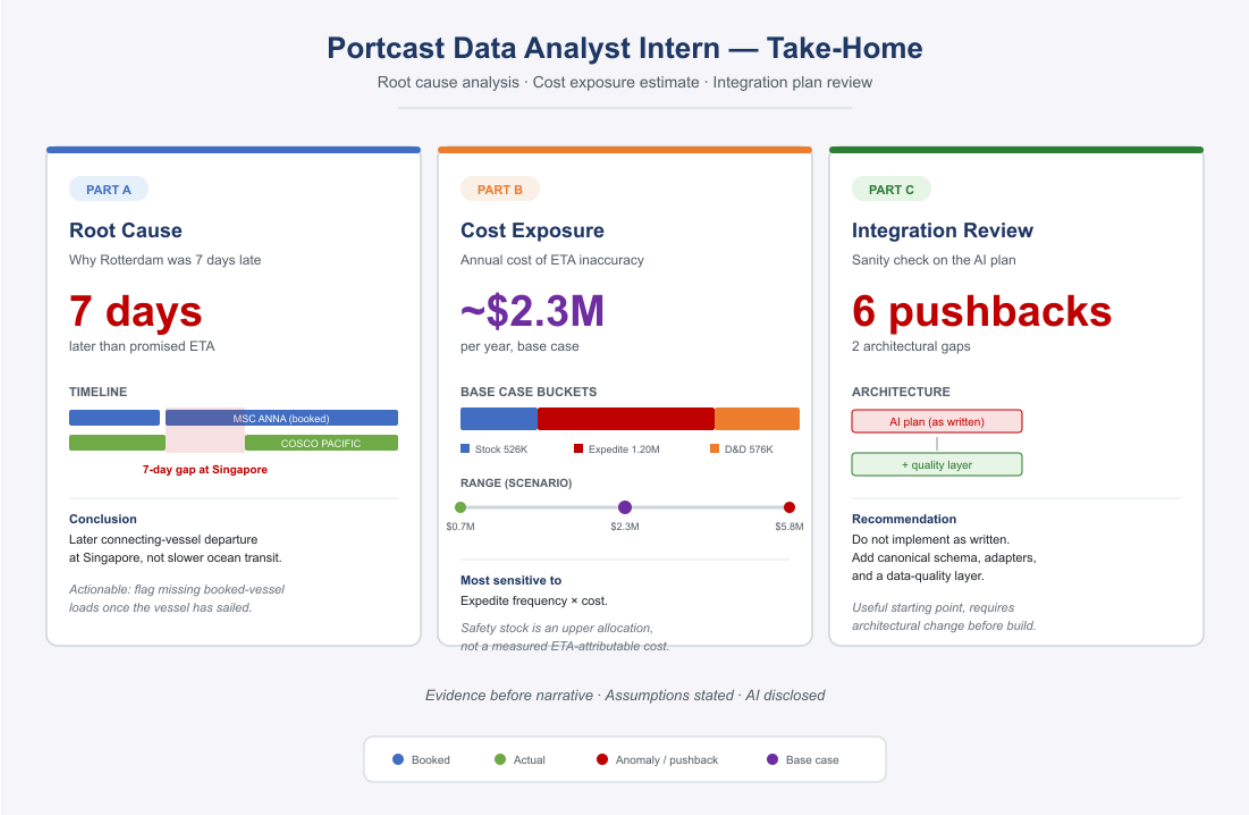
PORTCAST DATA ANALYST INTERN - TAKE HOME ASSIGNMENT



INTRODUCTION

This submission addresses three self-contained problems: a root cause analysis of a late container, an order of magnitude cost estimate for ETA inaccuracy, and a sanity check on an AI generated integration plan. Each part opens with the conclusion, followed by the evidence and reasoning. Where I have used AI, I have said so, specifically at the end.

PART A: ROOT CAUSE ANALYSIS

Conclusion: The 7 day ETA miss was caused by the shipment departing Singapore seven days later than planned. The container ultimately moved on COSCO PACIFIC rather than the booked MSC ANNA, consistent with a missed or rolled connection. **The supplied data does not establish why the booked connection was missed.**

ANOMALY

7-day gap at Singapore transshipment

Container discharged 3/10, loaded onto substitute vessel 3/17, departed 3/18.

The situation

A customer was quoted an ETA of 2024-04-15 into Rotterdam. The container actually arrived on 2024-04-22, 7 days late. The customer has heard "ocean delays" too often and no longer trusts it. I pulled the booked routing and the actual milestone events to determine the real cause.

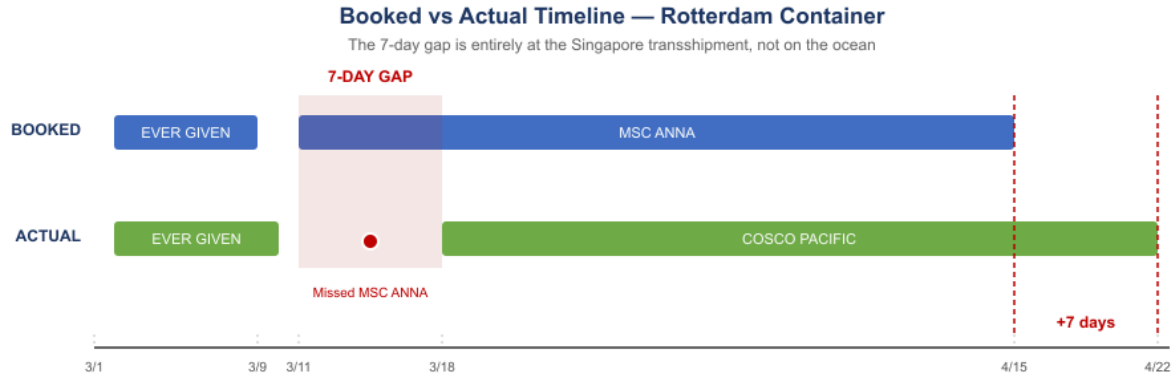
Data provided

Booked routing (as promised to the customer):

Leg	Route	Vessel (booked)	IMO	Planned departure	Planned arrival
1	Shanghai (CNSHA) → Singapore (SGSIN)	EVER GIVEN	9811000	2024-03-02	2024-03-09
2	Singapore (SGSIN) → Rotterdam (NLRTM)	MSC ANNA	9839430	2024-03-11	2024-04-15 (promised ETA)

Actual milestone events:

Milestone	Vessel (actual)	IMO	Timestamp
Gate-in, Shanghai	-	-	2024-03-01
Loaded, Shanghai	EVER GIVEN	9811000	2024-03-02
Discharged, Singapore	EVER GIVEN	9811000	2024-03-10
Loaded, Singapore	COSCO PACIFIC	9904455	2024-03-17
Departed, Singapore	COSCO PACIFIC	9904455	2024-03-18
Arrived, Rotterdam	COSCO PACIFIC	9904455	2024-04-22



Step 1: Decompose the delay

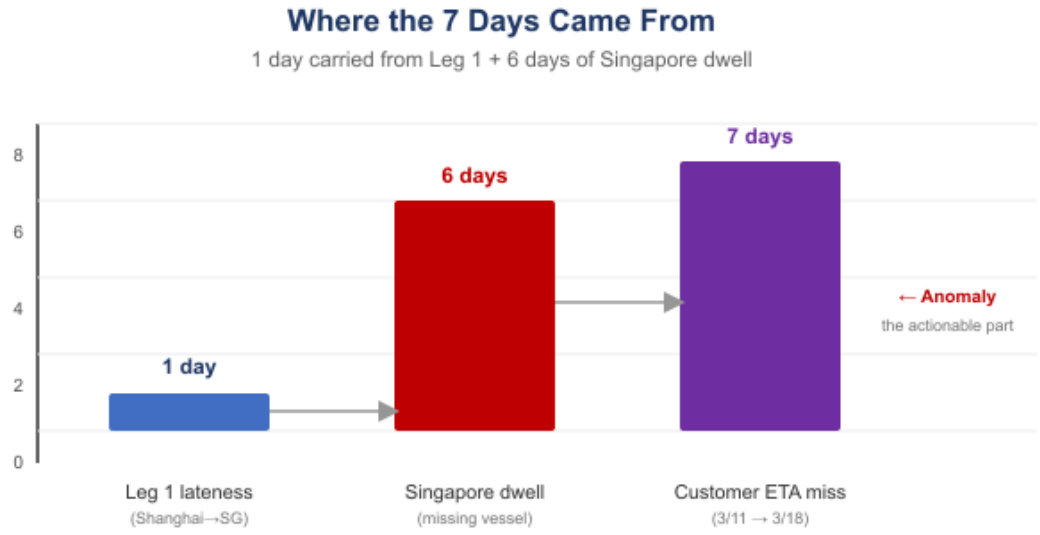
Leg 1 (Shanghai → Singapore): Booked arrival 3/9, actual discharge 3/10 → 1 day late.

Transshipment at Singapore: Booked connecting vessel MSC ANNA departing 3/11, actual connecting vessel COSCO PACIFIC departing 3/18 → 7 day later departure.

Leg 2 (Singapore → Rotterdam): Booked transit 3/11 - 4/15 = 35 days, actual transit 3/18 - 4/22 = 35 days → same transit time.

So the 7 day departure slip decomposes as roughly **1 day carried over from leg 1 plus 6 extra days of dwell in Singapore.** The planned connection window was about 2 days (3/9 to 3/11), leg 1's lateness cut that to roughly 1 day. That is a plausible reason the container missed MSC ANNA's cutoff. I would frame this as a hypothesis worth testing against carrier cutoff data, not as a proven cause - the supplied data does not contain MSC ANNA's loading cutoff or its own actual departure.

The planned and actual second leg transit times were both 35 days. Therefore, the final 7 day ETA miss is explained by the later second leg departure, with leg 1's 1 day lateness as a contributing factor at the connection.



Step 2: Identify the root cause

The data backed diagnosis is a later connecting vessel departure at Singapore. The container moved on COSCO PACIFIC rather than the booked MSC ANNA, consistent with a missed or rolled connection. The supplied data does not establish why the booked connection was missed.

The container was discharged from EVER GIVEN on 3/10, one day after the planned Singapore arrival. **The supplied milestone data contains no loaded event for the booked MSC ANNA.** Instead, the container was loaded onto COSCO PACIFIC on 3/17, which departed on 3/18. This is consistent with a vessel rollover: the supplied milestone data does not show the container moving on the booked connecting vessel, and it was subsequently loaded onto a later vessel.

Step 3: Rule out / assess alternatives

Ocean schedule slippage is not supported by the transit time evidence.

The actual second leg transit time is 35 days, identical to the booked 35-day transit. This means the seven day arrival delay is fully accounted for by the seven day later departure, rather than additional transit time on the Singapore to Rotterdam leg. That said, I do not have MSC ANNA's own actual departure from Singapore. If MSC ANNA also left late, that would be vessel level slippage rather than a rollover. Confirming whether MSC ANNA itself was late would require her AIS departure event.

Port congestion at Singapore is not established from the supplied data.

The data shows a 7 day later departure from Singapore, booked MSC ANNA ETD 3/11, actual COSCO PACIFIC ETD 3/18. The data does not establish whether congestion caused that gap. Congestion could have contributed to the rollover decision, but the supplied milestone events do not prove it. What the data does prove is that the container was loaded on a different vessel that departed later.

Step 4: Customer takeaway

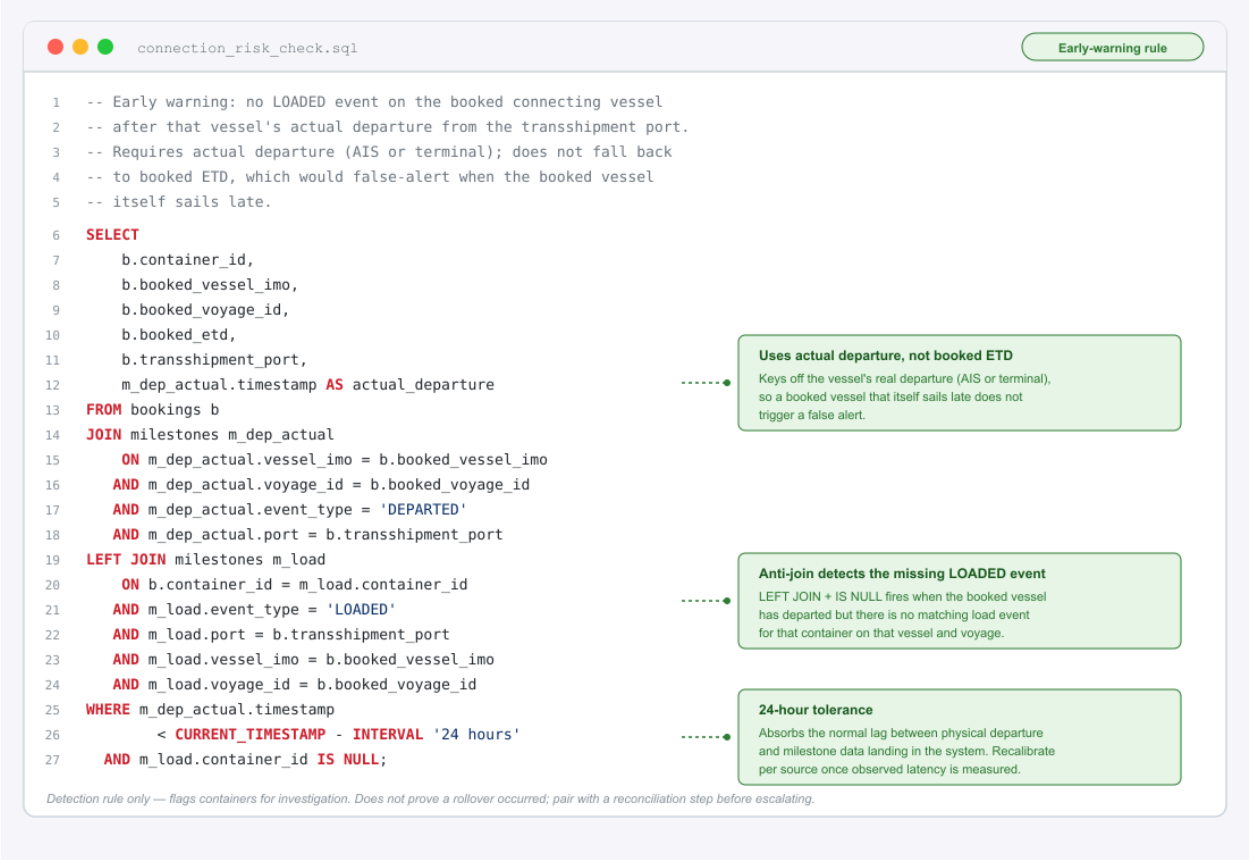
The customer should be told: this was a **later connecting vessel departure at Singapore, rather than slower transit on the Singapore to Rotterdam leg.** The milestone pattern is consistent with a missed connection onto MSC ANNA, although the supplied data does not establish why the connection was missed.

Assuming MSC ANNA sailed on schedule, the connection failure would have been visible in the tracking data around 3/11 - 3/12. From 3/12 to the promised ETA of 4/15 is 34 days, **roughly five weeks of warning time.** That is the payoff of building a monitoring signal, the problem was knowable early, and the customer could have been alerted with a revised ETA weeks in advance instead of being handed a post-hoc "ocean delays" explanation.

Step 5: Automated monitoring signal

A late firing rule that triggers only after the substitute vessel departs would be useless for customer communication. A better early warning rule, where actual departure coverage is available, is to **flag a container that has no LOADED event on its booked vessel once that vessel has actually departed,** with a tolerance to avoid false positives from data latency.

This is a connection-risk detection rule, not a root-cause rule. It identifies containers whose booked vessel has departed without a corresponding load event; a second-stage reconciliation step should determine whether the cause was rollover, data latency, missing milestones, or another exception.



Notes on the rule:

The query keys off the vessel's actual departure and joins on voyage ID, not just vessel IMO and port. Without the voyage key, a vessel calling Singapore multiple times would match every later call and produce duplicate rows and the wrong reference time.

The query does not fall back to the booked ETD. Falling back would reintroduce exactly the false-alert problem the rule is designed to avoid: a booked vessel that itself sails late would trigger an alert even though no connection has actually been missed.

Dependency: the query assumes the system contains an actual departure event for the booked vessel/voyage, from AIS or terminal data. If a carrier never publishes that departure event and no AIS coverage exists, the rule cannot fire for that carrier. That limitation should be tracked as a coverage metric per source, and carriers with poor departure event coverage should be flagged separately for manual review.

The 24-hour tolerance accounts for the lag between a vessel's actual departure and milestone data landing in the system.

The query assumes bookings are stored at leg level, so each leg has its own booked vessel, voyage, and ETD.

Two data quality risks need handling. First, many sources never emit a LOADED event, so a missing LOADED event is not proof of a missed connection, it can cause false positives. A corroborating signal available around the same time should be required before escalating, for example, the container still showing as discharged or in-yard at Singapore with no departure event, or a LOADED event on a different vessel at the same port. Second, LOADED events can be wrongly posted against the booked vessel, which causes false negatives. Cross-checking the vessel IMO and voyage ID against AIS position at the transshipment port mitigates this.

Assuming MSC ANNA sailed on schedule, this rule would have flagged the MSC ANNA → COSCO PACIFIC substitution around 3/12, giving **roughly 34 days of lead time before the promised ETA**. It is a detection rule, not proof of the operational reason, but it is a simple, high value QA check that aligns with Portcast's focus on anomaly detection and data quality.

PART B: GUESSTIMATE

Conclusion: Under illustrative assumptions, the **combined annual cost exposure across safety stock, expedited freight, and ETA-related detention/demurrage could be on the order of 0.7M - 5.8M USD, with a base case scenario of roughly 2.3M USD. The safety s tock component is an upper allocation rather than a measured ETA attributable cost.**

The question

A mid size manufacturer ships roughly 400 containers a month. Carrier ETAs drift by a few days versus reality. What is a reasonable order of magnitude estimate for the annual cost of inaccurate ETAs through safety stock, expedited freight, and port/warehouse detention or demurrage?

Step 1: Known number

400 containers/month × 12 = 4,800 containers/year.

Step 2: Cost buckets and flexed parameters

I use three buckets, with six flexed parameters across them. Not every ETA error creates cost, a one day drift may have zero financial consequence, so the percentages below are the share of shipments where drift actually converts into money.

Assumption 1 - Safety stock / inventory carrying cost (upper bound allocation). Average container value = 50,000 USD, annual inventory carrying cost = 20%. For an upper bound scenario, I allocate four days of the network's buffer inventory to ETA uncertainty. This is

intentionally conservative: in practice, only the portion of the buffer attributable to transportation uncertainty should be charged to inaccurate ETAs, and part of any safety buffer exists for demand variability, seasonality, and supplier variability. The figure below is therefore **a ceiling, not a point estimate.**

Assumption 2 - Expedited freight. 5% of shipments require expedited recovery, average incremental expedite cost = 5,000 USD.

Assumption 3 - Detention, demurrage, warehouse. 10% of shipments incur ETA related extra charges, average incremental cost = 1,200 USD.

Note on overlap: The expedite and D&D triggers can overlap, since one shipment could incur more than one type of ETA related cost. For this order of magnitude estimate, I calculate each bucket separately rather than attempting to model shipment level overlap. In a real analysis, I would join the cost data at shipment level to avoid double counting.

Step 3: Base-case calculation

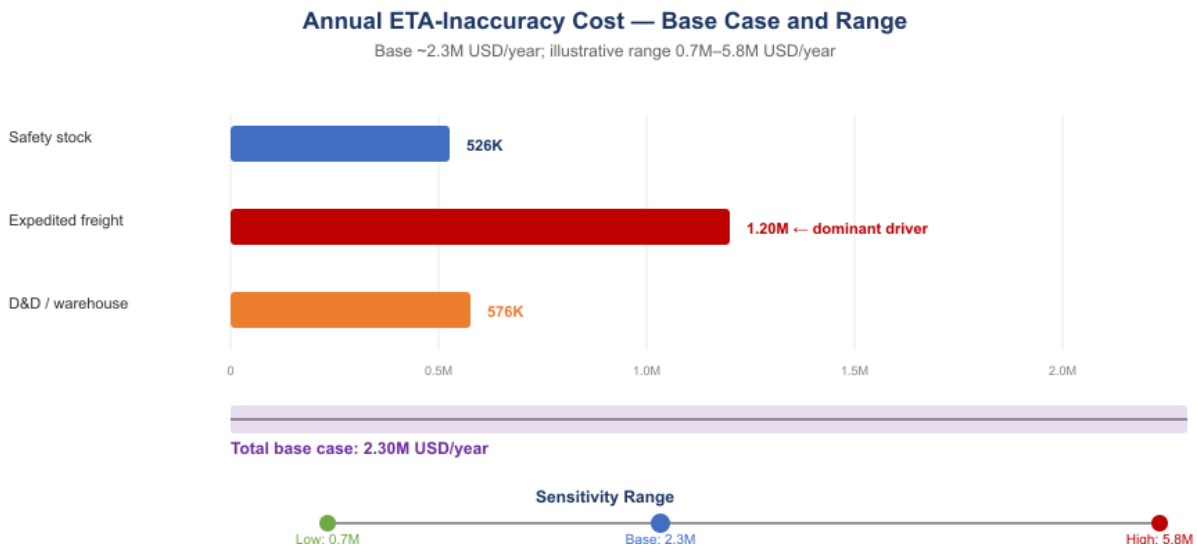
Safety stock (upper allocation): 4,800 × 50,000 × 20% × (4/365) ≈ 526,000 USD/year

Expedited freight: 5% of 4,800 = 240 shipments; 240 × 5,000 = 1,200,000 USD/year

Detention / demurrage / warehouse: 10% of 4,800 = 480 shipments; 480 × 1,200 = 576,000 USD/year

Step 4: Base-case total

Cost bucket	Annual estimate
Safety stock / inventory carrying cost (upper allocation)	~526K USD
Expedited freight	~1.20M USD
Detention / demurrage / warehouse	~576K USD
Total	~2.30M USD/year



Step 5: Range - low / base / high

Parameter	Low	Base	High
Buffer days held against uncertainty	2	4	6
Inventory carrying cost	15%	20%	25%
Expedite frequency	2%	5%	8%
Incremental expedite cost	3,000 USD	5,000 USD	8,000 USD
D&D frequency	5%	10%	20%
Incremental D&D cost	800 USD	1,200 USD	1,800 USD

Bucket	Low	Base	High
Safety stock (upper allocation)	~197K USD	~526K USD	~986K USD
Expedited freight	~288K USD	~1.20M USD	~3.07M USD
D&D / warehouse	~192K USD	~576K USD	~1.73M USD
Total	~0.7M USD	~2.3M USD	~5.8M USD

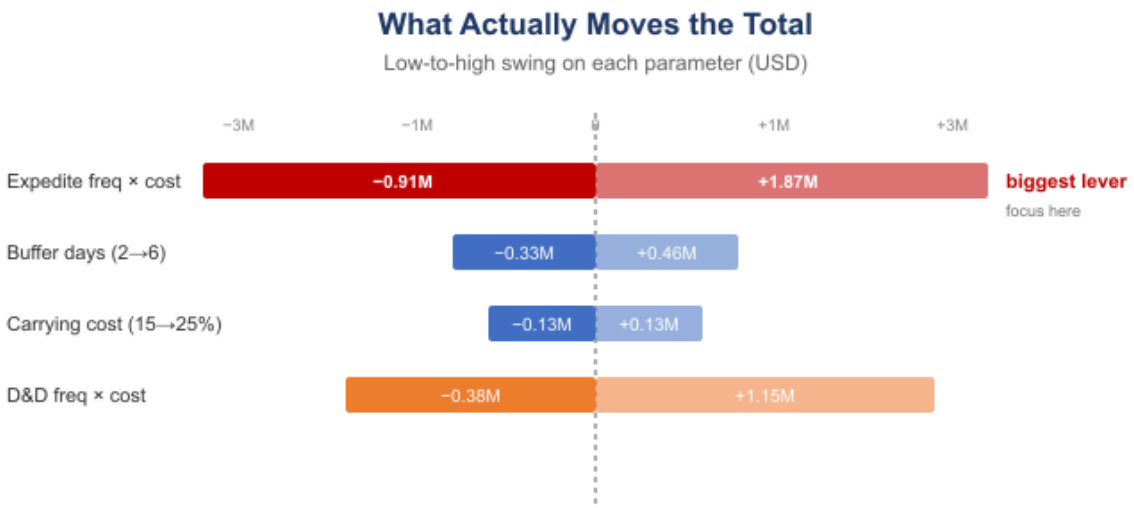
So the illustrative scenario range is roughly 0.7M - 6M USD/year, with a base case of about 2.3M USD/year. This is a scenario range, not a statistical confidence interval, and the safety-stock component within it is an upper allocation rather than a measured cost.

Step 6: Magnitude check

2.3M USD over 4,800 containers is about 480 USD per container, or under 1% of the roughly 240M USD annual cargo value. **This is a magnitude check, not independent validation of the estimate,** it establishes that the result is not obviously absurd relative to the size of the operation, nothing more. As a second cross-check, this is equivalent to holding about 2.6M USD of standing buffer inventory (240M USD annual throughput × 4/365) at a 20% carrying cost. One further caution: the share of shipments experiencing ETA drift is likely different from the 5 - 10% used here. The 5 - 10% figures represent the assumed share where drift converts into a measurable cost, not the share that experiences any drift at all.

Step 7: Sensitivity

The estimate is most sensitive to the percentage of shipments that require **expedited recovery and the average incremental expedite cost.** The full low to high swing on those two parameters alone is roughly 2.8M USD (from about 0.29M USD at the low end to about 3.07M USD at the high end), which is larger than the combined swing of every other bucket. The safety stock bucket is materially larger than it would be under a late shipment only allocation, but **the expedite bucket remains the dominant driver.**



Step 8: How to refine on the job

At Portcast, I would pull historical shipment data, join predicted ETA vs. actual arrival, and calculate the actual distribution of ETA error. Then I would correlate that error with expedite costs, D&D invoices, and inventory carrying costs. That would replace the illustrative assumptions with shipment level data and split the safety buffer by cause so only the ETA driven share is charged here.

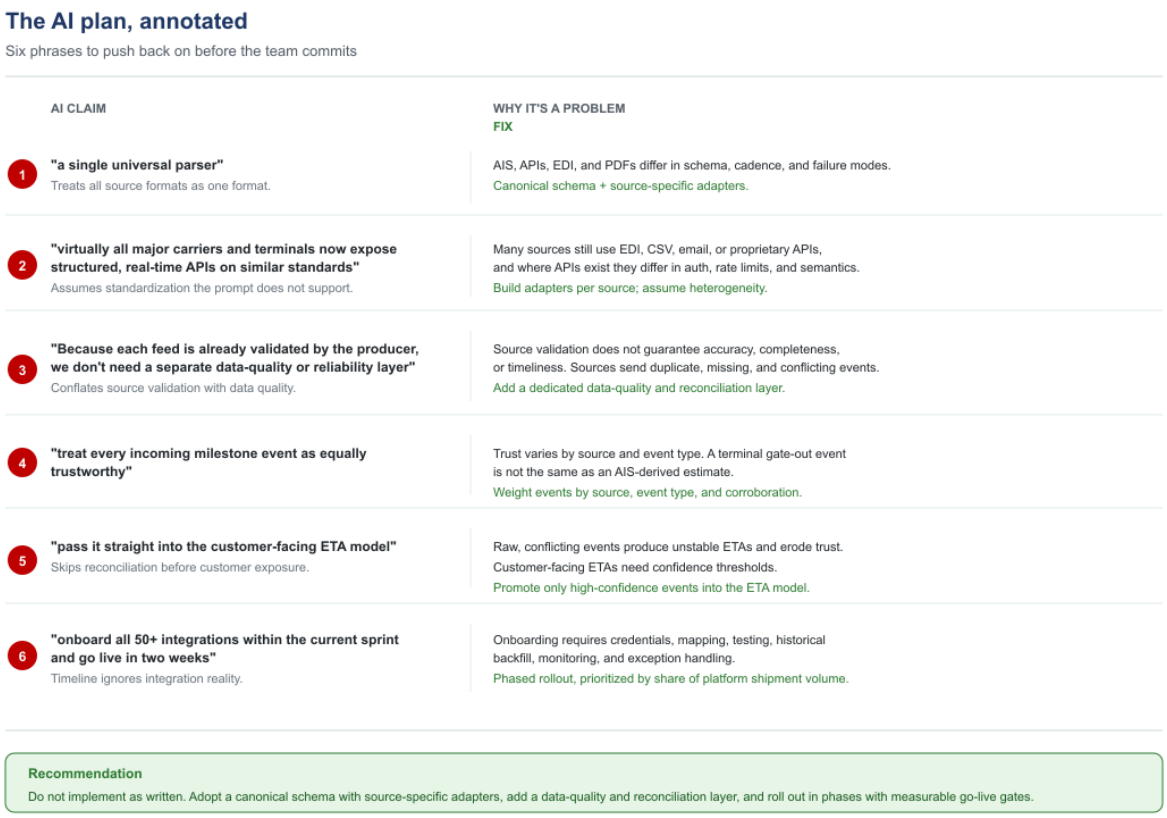
Note on assumptions: I do not have company-specific historical data, so I chose illustrative assumptions. In a real analysis, I would replace them with shipment-level ETA error and cost data.

PART C: AI OUTPUT SANITY CHECK

Conclusion: The proposed integration plan should not be implemented as written. It conflates fundamentally different data sources and lacks an independent data quality and reconciliation layer. It is a useful starting point, but **it requires architectural changes before implementation.**

The AI's paragraph

“To integrate carrier and terminal track-and-trace feeds at scale, we recommend building a single universal parser that maps every incoming feed - AIS pings, carrier APIs, terminal EDI messages, PDF schedules - into one common schema, since virtually all major carriers and terminals now expose structured, real-time APIs on similar standards. Because each feed is already validated by the carrier or terminal that produces it, we don't need a separate data-quality or reliability layer on our side - we can treat every incoming milestone event as equally trustworthy and pass it straight into the customer-facing ETA model. This lets us onboard all 50+ integrations within the current sprint and go-live in two weeks.”



Claims/decisions I would push back on

1. "Build a single universal parser for every feed."

Simplifies a heterogeneous set of source formats. AIS pings, carrier APIs, terminal EDI, and PDF schedules differ in schema, cadence, and failure modes. A common canonical schema with source-specific adapters is more durable than a single parser.

2. "Virtually all major carriers and terminals now expose structured, real-time APIs on similar standards."

Assumes more API standardization than the information provided supports. Many sources still use EDI, CSV, email, or proprietary APIs, and where APIs exist they differ in auth, rate limits, and event definitions.

3. "Because each feed is already validated by the producer, we don't need a separate data-quality or reliability layer on our side."

Omits a data quality layer that source behavior will likely require. Source validation does not guarantee accuracy, completeness, or timeliness, and sources send duplicate, missing, late, and conflicting events.

4. "Treat every incoming milestone event as equally trustworthy."

Trust should vary by source and event type. A terminal gate out event may be more reliable than an AIS-derived estimate, and events from different sources should be reconciled rather than blended blindly.

5. "Pass every event straight into the customer-facing ETA model."

Customer-facing ETAs need reconciliation and confidence thresholds. Raw, conflicting events can produce unstable or inaccurate ETAs.

6. "Onboard all 50+ integrations within the current sprint and go live in two weeks."

Not supported by the information provided. Onboarding requires credentials, mapping, testing, historical backfill, monitoring, and exception handling. A phased rollout is more realistic.

Two sharper technical points

AIS pings are vessel-level, not container-level. AIS tells you where a ship is, not where a specific container is. To use AIS in a container ETA model, you need a mapping from container to vessel to voyage, and that mapping is itself derived from carrier and terminal events. The plan treats AIS as if it were a container milestone, it is not.

PDF schedules are planned data, not milestones. Carrier PDFs and proforma schedules describe what is supposed to happen. Milestone events describe what did happen. Mixing them

into one "common schema" without distinguishing planned, estimated, and actual event types will cause the ETA model to confuse its own inputs. The canonical schema needs an explicit event-class field.

Rewrite of the two most important sentences

Original: "To integrate carrier and terminal track-and-trace feeds at scale, we recommend building a single universal parser that maps every incoming feed - AIS pings, carrier APIs, terminal EDI messages, PDF schedules - into one common schema…"

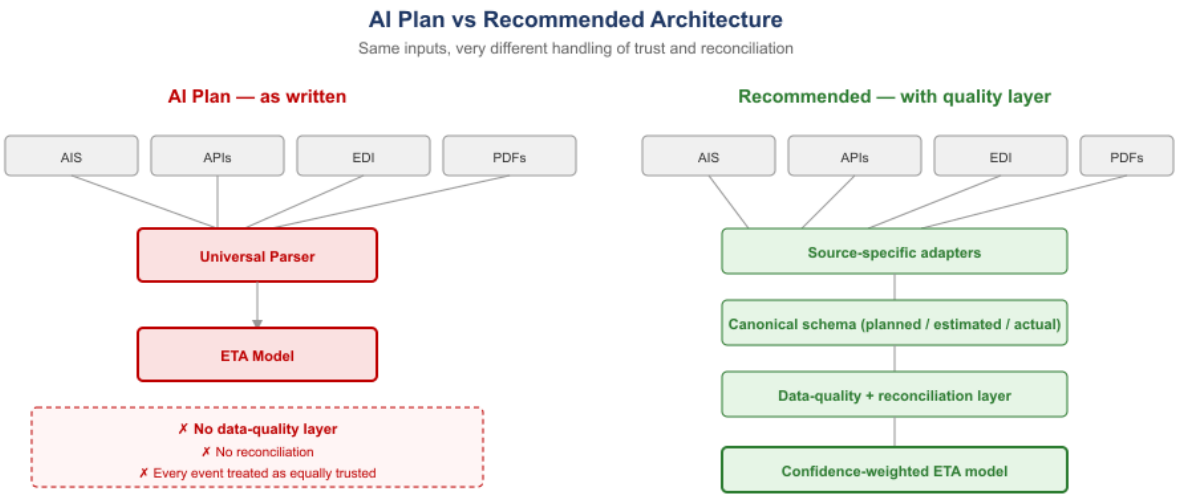
Rewrite: "To integrate carrier and terminal track-and-trace feeds at scale, build **a common canonical event schema with source specific adapters** and parsers for APIs, EDI, AIS, and PDFs. The schema must distinguish planned, estimated, and actual events, and AIS must be joined via a container to vessel mapping rather than treated as a container-level milestone."

Original: "Because each feed is already validated by the carrier or terminal that produces it, we don't need a separate data-quality or reliability layer on our side - we can treat every incoming milestone event as equally trustworthy and pass it straight into the customer-facing ETA model."

Rewrite: "Because source validation does not guarantee accuracy or completeness, we need **a separate data-quality and reliability layer that deduplicates, reconciles conflicting milestones, and only promotes validated events** into the customer-facing ETA model."

A better plan in one paragraph

A more realistic plan would be built in stages. First, define a **canonical event schema that explicitly separates planned, estimated, and actual events.** Second, build source-specific adapters rather than one universal parser, and stand up a data quality layer for deduplication and reconciliation with per-source trust weightings. Third, roll out in phases, prioritizing carriers by share of platform shipment volume, then measure how much volume is covered by the first five before setting the rollout cutoff. Fourth, attach measurable go-live gates to each phase, using as **illustrative starting thresholds** an event match rate of at least 95%, P95 milestone latency no greater than 24 hours, and a duplicate event rate no higher than 1%. The match rate metric should be defined explicitly at the outset, for example, same container, same event type, same port, and timestamps within a defined window. All three thresholds should be **recalibrated against observed source performance** during the first two or three carrier onboardings. Fifth, build monitoring and alerting from day one. This aligns with Portcast's focus on clean, connected, reliable data.



AI VS. MY OWN EFFORT

I drafted each part myself, then ran the draft through an AI review and decided which changes to adopt. I treated the AI review as a critique rather than an answer generator, I evaluated each suggestion, accepted only changes I could justify, and rewrote the final reasoning myself.

Part A: I built the initial timeline decomposition and the transit time comparison myself. The leg 1 carry over decomposition (1 day + 6 days) and the caveat about MSC ANNA's own departure came from the AI review, I agreed with both and incorporated them. I used AI to sanity check my first SQL query, and the review flagged that it fired too late and that the fallback to booked ETD would false alert on late sailing vessels. I agreed with both critiques and rewrote the query. The voyage-ID join, the removal of the booked ETD fallback, and the explicit dependency note on departure event coverage came from follow up review rounds.

Part B: I built the initial cost model and did the arithmetic myself. The AI review flagged that I had applied the safety stock buffer only to the 10% of shipments that ran late, which understated that bucket roughly tenfold, I agreed and corrected it. The low/base/high table and the sanity check were suggested in review, I chose the flex ranges and wrote the surrounding justification. The framing of the safety stock figure as an upper allocation rather than a measured cost came from a later review round.

Part C: I wrote the initial pushback list and the two rewrites myself. The AI review prompted me to lead with an explicit recommendation and to fold the better plan back into paragraph form, both of which I agreed with and did. The two sharper technical points about AIS and PDF schedules came from review, and I incorporated them into the rewrites. The go-live gates came from review, I chose the specific thresholds and the phasing logic. The softening of the carrier-volume claim, the definition of the match-rate metric, and the tone adjustments in the pushback list came from follow up review rounds.